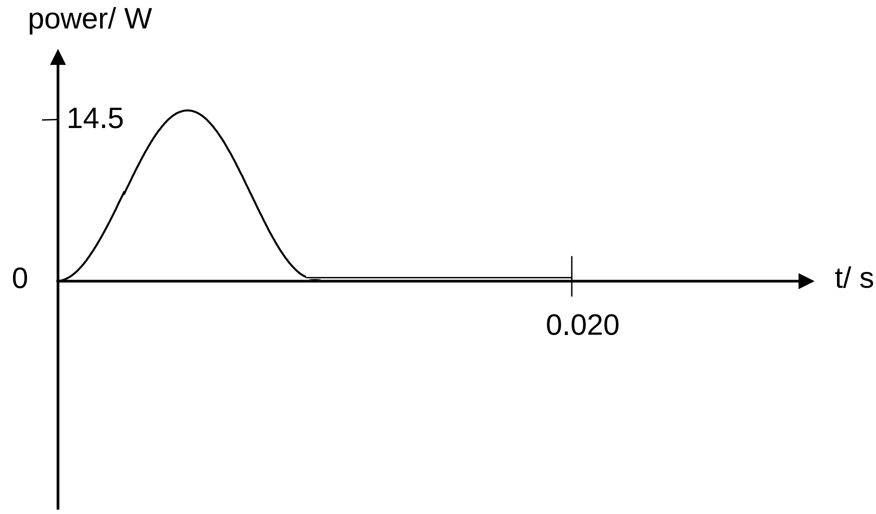
Qns	Answer	Marks
3(a)	r.m.s. value of an alternating current is the value of a steady direct current that will dissipate thermal energy at the same average rate as the a.c. in a given resistor	**A1**
3(b)	peak voltage = $6.0\sqrt{2}$ peak voltage = 8.5 V	 **A1**
3(c)	$k = \omega = 2\pi f = 2\pi(50) = 310$ Unit: radian per second, rad s^{-1} (or s^{-1})	**A1** **A1**
3(d)	8.5 V	**A1**
3(e)	zero because either no current in circuit (and V = IR) or all p.d. across diode	**A1**
3(f)	 $P_{\max} = \frac{V_o^2}{R} = \frac{8.5^2}{5.0} = 14.5$ W $T = \frac{1}{f} = \frac{1}{50} = 0.020$ s Shape and showing one complete cycle Labelling max power as 14.5 W Labelling period as 0.020 s	**M1** **A1** **A1**

Qns	Answer	Marks
4(a)(i)	A beam of white light consists of photons of all wavelengths (λ) in the visible	